1

$$\bar{x} \in \left(\mu - 1.96 \times \frac{\sigma}{\sqrt{n}}, \mu + 1.96 \times \frac{\sigma}{\sqrt{n}}\right)$$

is equivalent to

$$\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \in (-1.96, +1.96)$$

and ~~or~~ also equivalent to

$$\frac{X_1 + \dots + X_n - n\mu}{\sqrt{n}\,\sigma} \in (-1.96, +1.96)$$

that is approximately insured ~~by~~ the Central Limit Theorem.

Adding up independent random variables

The density function of $X+Y$ (if X and Y are independent) is

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(t)\, f_Y(z-t)\, dt$$

This is called the convolution of the individual density functions.

Example: for $X \sim U[0,1]$ and $Y \sim U[0,1]$
(X and Y are uniformly distributed on $[0,1]$ interval)

$$f_{X+Y}(z) = \int_0^z 1 \cdot 1 \, dt = z \quad \text{if} \quad 0 \le z \le 1$$

$$\text{and} = \int_{z-1}^{1} 1 \cdot 1 \, dt = 2-z \quad \text{if} \quad 1 < z \le 2$$

because in the first case, $\underbrace{0 \le t \le 1,}$ and

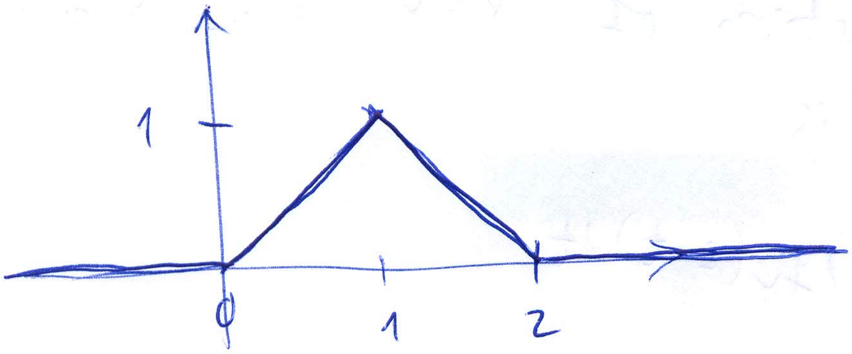
$$0 \le z - t \le 1$$
$$-z \le -t \le 1-z$$
$$\underbrace{z \ge t \ge z-1,}$$

imply

$$\boxed{0 \le t \le z},$$

and in the second case, the same two conditions imply $\boxed{z-1 \le t \le 1}$.

So the density function of $X+Y$ looks like this.



<u>Example</u>: Calculate the density function of the sum of 3 independent random variables, each having $U[0,1]$ distribution.

We already know that $f_{X+Y}(z) = \begin{cases} z, & \text{if } 0 \leq z \leq 1 \\ 2-z, & \text{if } 1 \leq z \leq 2 \\ 0 & \text{otherwise}. \end{cases}$

Now $X+Y+Z$ will have values between 0 and 3.

$$f_{X+Y+Z}(v) = \int_{-\infty}^{\infty} f_{X+Y}(t)\, f_Z(v-t)\, dt,$$ with different rules

for 3 cases: a) $0 \leq v < 1$
b) $1 \leq v < 2$
c) $2 \leq v \leq 3$.

We always have $0 \leq v-t \leq 1$, that is $-v \leq -t \leq 1-v$, $\boxed{v \geq t \geq v-1}$.
and also $\boxed{0 \leq t \leq 2}$.

This implies: ~~E.g. in case~~ a) $0 \leq t \leq v$
b) $v-1 \leq t \leq v$
c) $v-1 \leq t \leq 2$

Please finish finding $f_{X+Y+Z}(v)$ in each case!